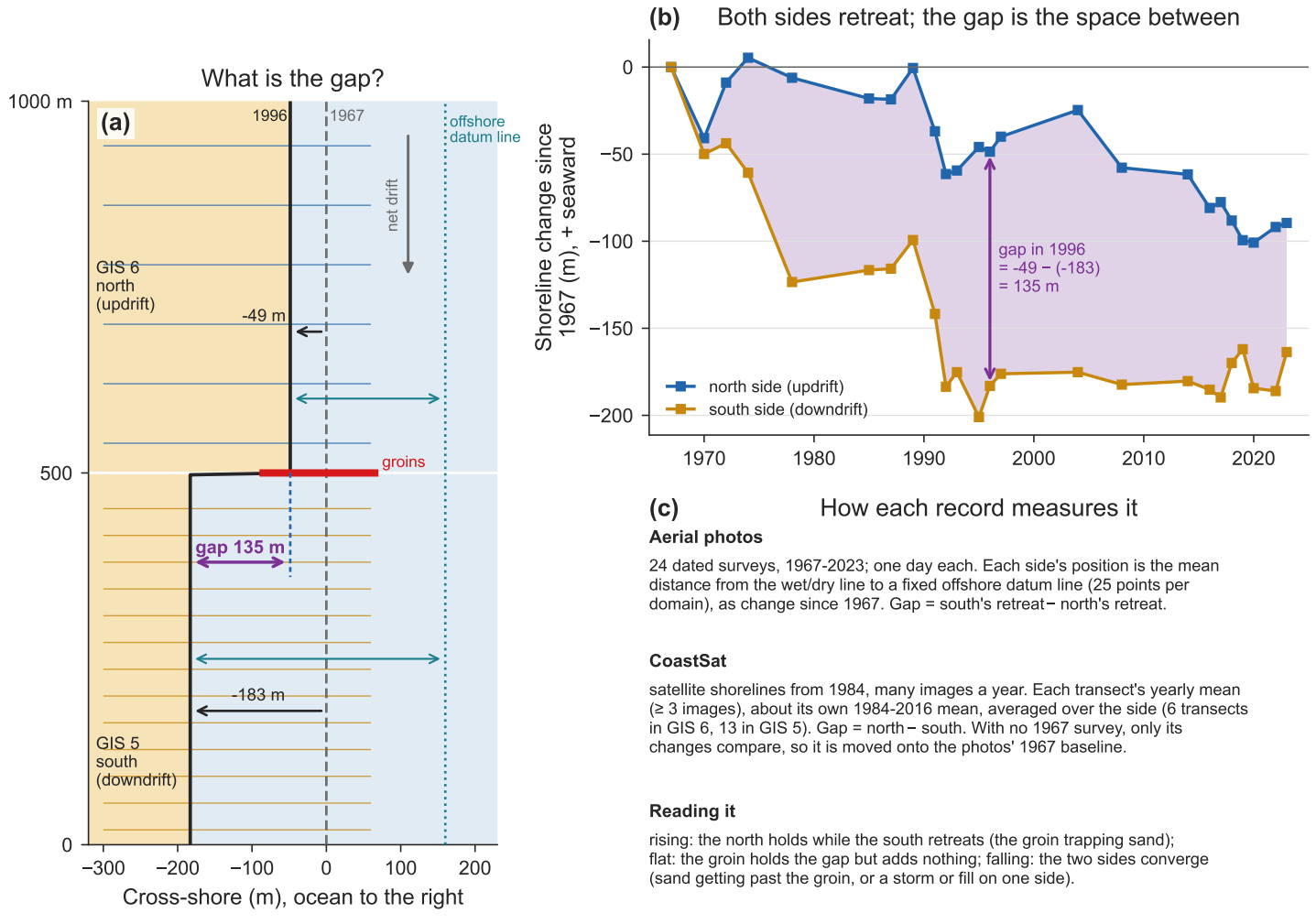
What is the gap?
(a)
1996
1967
offshore datum line
net drift
GIS 6 north (updrift)
-49 m
groins
gap 135 m
-183 m
GIS 5 south (downdrift)
1000 m
500
0
−300
−200
−100
0
100
200
Cross-shore (m), ocean to the right
(b) Both sides retreat; the gap is the space between
Shoreline change since 1967 (m), + seaward
0
−50
−100
−150
−200
gap in 1996 = -49 − (-183) = 135 m
north side (updrift)
south side (downdrift)
1970
1980
1990
2000
2010
2020
(c) How each record measures it
Aerial photos
24 dated surveys, 1967-2023; one day each. Each side's position is the mean distance from the wet/dry line to a fixed offshore datum line (25 points per domain), as change since 1967. Gap = south's retreat − north's retreat.
CoastSat
satellite shorelines from 1984, many images a year. Each transect's yearly mean (≥ 3 images), about its own 1984-2016 mean, averaged over the side (6 transects in GIS 6, 13 in GIS 5). Gap = north − south. With no 1967 survey, only its changes compare, so it is moved onto the photos' 1967 baseline.
Reading it
rising: the north holds while the south retreats (the groin trapping sand);
flat: the groin holds the gap but adds nothing; falling: the two sides converge (sand getting past the groin, or a storm or fill on one side).
1967 shoreline
1996 shoreline (photos)
groins (simplified)
CoastSat transects, north
CoastSat transects, south
photo distance to the datum